

Real-Time Human Detection and Tracking

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Abstract—Real-time human detection and tracking is a vast, challenging and important field of research. It has wide range of applications in human recognition, human computer interaction (HCI), video surveillance etc. The research for biometric authentication of a person has reached far but the real-time tracking of human beings has not gained much importance. Tracking of human being can be used as a prior step in biometric face recognition. Keeping continuous track of person will allow to identify person at any time. The system consist of two parts first human detection and secondly tracking. Human detection step is split into face detection and eye detection. Face is a vital part of human being represent most important information about the individual. Eyes are the important biometric feature used in person identification. Face detection is done using skin color based methods. $YCbCr$ color model is used to detect skin regions as it represents intensity and color information separately. For eye region detection projection function and pixel count methods are used. Finally the mean shift algorithm and kalman filter algorithm is used for tracking.

Keywords—Face Detection, Eye Detection, Projection Function, Kalman Filter, Mean Shift Algorithm

I. INTRODUCTION

Automatic human detection and tracking is an important and challenging field of research and has many application areas. The increasing use of computer vision in surveillance, replacing human beings, has initiated the research in the field of face detection. Early research is biased to human recognition rather than tracking. Monitoring the movements of human being raised the need for tracking. Monitoring movements are of high interest in determining the activities of a person and knowing the attention of person.

The method proposed in here is divided into three steps: face detection, eye detection and tracking. Firstly face is detected in frame. Secondly the eye region is located in face region. Finally co-ordinates of eye region are calculated and tracked. The block diagram of the system is shown in Fig. 1.

Face is important part of human anatomy which varies from individual to individual but can be located with the use of certain context information. Facial feature detection methods are classified into two: local and global methods. In local method each face component like eyes, mouth and nose is detected separately while in global method all facial parts are detected jointly and a model based on relative position of these features is constructed. The goal of face detection method is to determine the presence of face in the frame and return its location. Numerous techniques have been developed to detect faces in images. The simplest technique used for face detection is based on skin color segmentation which requires less computations. The RGB image is converted into $YCbCr$ or HSV color space and threshold is applied to detect

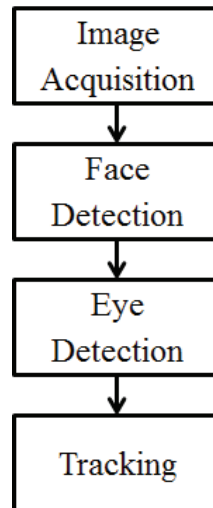


Fig. 1. Block diagram of system

face region. There are many other methods that are based on geometric features of face but these methods are complex and requires more computation time. The starting point for any face detection technique, is detecting faces in a single image. After developing an algorithm which works successfully on still images is evaluated on the video streams taken from a camera or other source.

Eye is the most complex organ in human body. The retina is a membrane containing photoreceptor nerve cells that lines the inside back wall of the eye. The photoreceptor nerve cells of retina changes the light rays entering through pupil into electrical impulses and transmit them through optic nerve to the visual part of brain where an image is perceived. Image analysis is used to define and understand the information contain of image by the computer vision applications. Eye tracking and eye movement-based interaction using computer vision techniques have important role in future perceptual user interfaces. Eyes are detected using projections of face in horizontal and vertical directions.

Tracking is the most challenging step of tracking system which localizes and associates the feature across series of frames. Tracking can be done by detecting the object in each frame but it requires large time that affects the frame rate of the system. To reduce detection time, probable search area for the object in next frame is predicted based on the location of that object in present frame. The two major components of tracking system are object representation and data association. The first component object representation, describes the object and second, data association utilizes the information about target object to track it in each subsequent frame. After detecting

desired feature of interest (face & eyes) in the image its locations are computed and passed to tracking algorithm. The algorithm used for tracking is based on Dorin Comaniciu and Visvanathan Ramesh [1] where mean shift analysis is used to predict and find location of object in next frame. The object description is done using color histogram bins and the most similar region is found in next frame around the location of object region in previous frame.

II. RELATED WORK

Vast research carried out in last decade has led to many new technologies and applications of image processing.

Human skin color ranges from very dark to pinkish-white. The human skin color has two important properties- human skin colors cluster in a small regions and skin colors differ more in brightness than in colors. Color distribution in an image is represented using color models like - RGB , YC_bC_r , HSV etc. RGB color space represents color of each pixel as the combination of R , G and B components, where each components value lies in the range 0 to 255. The color information is very sensitive to the intensity, therefore color component values are normalized with intensity I and expressed by 2D Gaussian distribution [2]. The pixel is classified as skin pixel by using r and g values where r value is in the range 0.36 to 0.456 and g value is in the range 0.28 to 0.363. The disadvantage of RGB model is that it is not perceptually uniform and does not have separate intensity and color components.

In HSV model $Hue(H)$ is a measure of the spectral composition of a color and represented as an angle, which varies from 0 to 360° . $Saturation(S)$ refers to the purity of colors and $Value(V)$ is the intensity of pixel, which ranges from 0 to 1 [3]. Yang et al.[4] thresholded H plane to detect skin pixels. Chae et al. [5] constructed statistical color model using Baye's probability theory and assigned membership function (MF) to each pixel.

The mostly used color space is YC_bC_r where Y is luminance component, C_b is blue chrominance and C_r is red chrominance. The separate luminance and chrominance component makes this model illumination invariant. Skin pixel has C_r value near to 100 and C_b value around 150. Kumar et al. [6], removed intensity and blue chrominance component and remodeled image back to RGB space as skin color is influenced by red color of blood.

Combination of more than one model for face detection is presented in [7] where combination of HSV model and YC_bC_r model is used. The range for classifying skin pixel is $Cb \leq 125$ or $Cb \geq 160$, $Cr \leq 100$ or $Cr \geq 135$, $26 < Hue < 220$. Srisuk et al. [8], first transformed image into chrominance space and then using the Bayes probability distribution the skin and non-skin pixels were categorized. Saber et al.[9] introduced YES color space where Y is intensity component and E & S chroma components obtained by shifting bits in RGB . Table I gives summary of face detection using skin color.

Eye is an important biometric feature of human anatomy and popularly used for human recognition. Eye tracking is most often used for human activity recognition, estimate gaze

TABLE I. SUMMARY OF COLOR MODELS

Author	Model	Criteria
Sirohey et al.[10]	HSV	$0 < H < 50, 0.20 < S < 0.68, 0.35 < V < 1$
Wang et al.[11]	RGB	$0.36 \leq 0.456, 0.28 \leq g \leq 0.363$
Vranceanu [12]	HSV	$0.05 < H < 0.2, 0.15 < S < 1, 0.3 < V < 1$
Tang et al. [13]	YC _b Cr	$90 < C_r < 125, 135 < C_b < 165$
Yap et al. [7]	YC _b Cr	$100 < C_r < 135, 125 < C_b < 160, H < 226$
Vranceanu [12]	YC _b Cr	$Y > 90, 85 < C_b < 135, 135 < C_r < 180$
Huang et al.[14]	YC _b Cr	$80 < C_r < 120, 133 < C_b < 165$
Ming et al.[15]	YC _b Cr	$80 \leq C_r \leq 127, 137 \leq C_b \leq 165$

direction and human thought processes. The intensity pattern near eye region changes drastically than rest of face region. Eye region can be detected by taking projections of face region in horizontal and vertical directions.

A projection of image in horizontal or vertical direction is sum of pixels of each row or column respectively. Projection function finds boundary between different image regions. The boundary region is signified by rapid changes in the projection function. Eye detection based on simple gray projection is presented in [16]. In order to locate eyes better and filter the interference in the curve, 1D discrete wavelet transform is used and required low-frequency curve is obtained [17].

Tracking aims at detecting location of object in next frame based on location of the object in current frame. Kalman filter estimates position and uncertainty of moving targets in next time frame, i.e. where to search for target in next frame. Kalman filter estimates the new location of target using the difference of the movement of the target within last two frames [18]. Mean shift tracking is an appearance based object tracking method. It employs mean shift analysis to identify a target candidate region which has most similar appearance to the target model in terms of intensity distribution [19].

III. FACE DETECTION METHODS

Automatic face detection is first step in many applications like face recognition, face retrieval, face tracking, surveillance, HCI, etc. Developing a model of face is difficult because face is complex, multi-dimensional visual stimuli. Human face changes with internal factors like facial expression, beard, moustache, glasses and external factors like scale, lightening conditions and orientation. The changes in skin color are not much compared to the above mentioned factors and can be used to locate human being. HSV and YC_bC_r color models are mostly used for face detection.

A. HSV Model

HSV coordinate system is cylindrical where vertical position defines brightness, angular position defines hue and radial position defines saturation. HSV color model is alternative to the classical color spaces such as RGB because it is similar to the way humans tend to recognize color. Since most of the digital images processing systems operate on RGB images, the conversion from RGB to HSV is done using following

equations:

$$H_1 = \cos^{-1} \frac{0.5 [(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}} \quad (1)$$

$$H = \begin{cases} H_1 & \text{if } B \leq G \\ 360^\circ - H_1 & \text{if } B \geq G \end{cases} \quad (2)$$

$$S = \frac{(Max(R, G, B) - Min(R, G, B))}{(Max(R, G, B))} \quad (3)$$

$$V = \frac{(Max(R, G, B))}{255} \quad (4)$$

The pixel is classified as skin pixel if it has values of H and S given by Equation 5.

$$0 \leq H \leq 0.2, 0.2 \leq S \leq 0.7, \quad (5)$$

The values of threshold remains almost same for all variations of human skin color.

B. YC_bC_r Model

The mostly used color space is YC_bC_r where Y is luminance component, C_b is blue chrominance and C_r is red chrominance. The conversion of image from RGB to YC_bC_r is done using following equations:

$$Y = 0.299R + 0.587G + 0.114B \quad (6)$$

$$Cb = 128 + (-0.169R + 0.33G + 0.5B) \quad (7)$$

$$Cr = 128 + (0.5R - 0.419G - 0.081B) \quad (8)$$

Skin pixel has C_r value around 100 and C_b value around 150 thus pixel is classified by selecting proper range of threshold given in Equation 9.

$$(R, G, B) = \begin{cases} 255, & \text{if } Cr \in [80, 120] \text{ and } Cb \in [135, 165] \\ 0, & \text{if } Cr \notin [80, 120] \text{ and } Cb \notin [135, 165] \end{cases} \quad (9)$$

Thresholding detects skin regions in frame and morphological operations are utilized to locate face accurately.

C. Morphological Processing

Morphological operation is a tool for extracting image components that are useful in representation and description of region shape. The two important morphological operations are erosion and dilation shown in Fig. 2. These operations are based on structuring element - a small set or sub images used to probe an image under study for properties of interest.

If A and B are sets in Z^2 , the erosion of set A by structuring element B is given by Equation 10.

$$A \ominus B = \{z | (B)_z \subseteq A\} \quad (10)$$

Create a new set by running B over A so that origin of B visits every element of A . At each location of origin of B if B is completely contained in A mark that location as member of new set else mark it as not a member of new set.

If A and B are sets in Z^2 , the dilation of set A by structuring element B is given by Equation 11.

$$A \oplus B = \{z | (B)_z \cap A \neq \phi\} \quad (11)$$



(a) Binary Image (b) Dilation (c) Erosion

Fig. 2. Morphological operations

Create a new set by running B over A so that origin of B visits every element of A . At each location of origin of B , if B and A overlap by at least one element, mark that location as member of new set else mark it as not a member of new set.

IV. EYE DETECTION METHODS

The shape of the eye is elongated with the long axis horizontal. Intensity values between flesh region of face and sclera has large difference. Eye region has high gray variation with respect to other face regions. The approach adopted for eye region detection is based on gray image integral projection and pixel count.

A. Integral Projection Method

The 3D face image consists of valleys for eye regions. When the 3D image is presented in 2D space the eye region appears dark compared to other face region. In gray image these regions are discriminated using their lower values. A projection of image in horizontal or vertical direction is used to detect the boundary between different image regions [20]. Horizontal and vertical projection of an image is given by Equation 12 and 13 respectively.

$$HP_x = \sum_{y=1}^{N_{Columns}} f(x, y) \quad (12)$$

$$VP_y = \sum_{x=1}^{N_{Rows}} f(x, y) \quad (13)$$

The region above the eyebrows and below the eyes is brighter and the uppermost valley in the horizontal projection function corresponds to eyes. This region starts above the eyebrows and extends to the region below the eyelids, and includes two eyebrows, two eyes, eyelids and eye corners. Similarly in vertical position two valleys corresponds to the position of two eyes. The projects of face is shown in Fig. 3.

B. Pixel Count Method

Eyes lie in the upper half of face, it can be detected using the face geometry. In pixel count method a window is defined and face image is scanned. At each location the numbers of black pixels are measured. The window is moved over entire face image with single pixel increment. The window with maximum number of black pixel count corresponds to the eyes. The algorithm is as follows:



Fig. 3. Horizontal projections

Algorithm 1 Pixel Count

- 1: Convert face image to gray scale.
 - 2: Perform histogram equalization to normalize intensity.
 - 3: Convert gray scale image to binary image.
 - 4: Select a window whose width is equal to face width and height 20% of face height.
 - 5: Scan face image from top to bottom and count number of dark pixels in window.
 - 6: Find window with maximum number of dark pixels.
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The window with the maximum number of dark pixel corresponds to eye region because the eye region contains eye brows and retina which forms black pixels in binary image.

V. TRACKING

Tracking objects in video sequences is one of the most important tasks in computer vision. The approach is to search target within a region centered on last position of the object. The three main facts that should be taken care for effective tracking are: speed of moving object, frame rate of video and search region. These factors are dependent on each other, higher frame rate is used to get hold of faster moving target which on other hand requires faster processing. Kalman filter and mean shift analysis used for tracking is described below.

A. Kalman Filtering

Kalman filter predicts the next time state vector based on the movement information in the previous stages [18]. Kalman filter equations are of two types: time update equations and measurement update equations. The time update equations projects forward the current state and error covariance estimates to obtain the apriori estimates for the next time step. The measurement update equations are the feedback i.e. it incorporates a new measurement into the a priori estimate to obtain an improved aposteriori estimate. The time update equations are predictor equations, while the measurement update equations are corrector equations. The process for Kalman filter estimation and update is given in Algorithm 2.

The state equation is

$$X_k = AX_{k-1} + Bw_k \quad (14)$$

Algorithm 2 Kalman Filter

- 1: Initial Estimates for $\hat{X}_{k-1}$ and P_{k-1}
- 2: Time Update

- Project the state

$$\hat{X}_k^- = A\hat{X}_{k-1} + Bw_k$$

- Project the error covariance

$$P_k^- = AP_{k-1}A^T + Q$$

- 3: Measurement Update

- Compute Kalman gain

$$K_k = \frac{P_k^- H^T}{H P_k^- H^T + R}$$

- Update estimate with measurement Z_k

$$\hat{X}_k = \hat{X}_k^- + K_k(z_k - H\hat{X}_k^-)$$

- Update error covariance

$$P_k = (1 - K_k H) P_k^-$$

The measurement model is

$$Z_k = HX_k + v_k \quad (15)$$

A and B are the system parameters and are matrix in the multi-model system, H is a parameter in a measure system, which is matrix in the multi-measurement system. w_k and v_k represents the process and measurement noise respectively.

The state vector $X_k = [x_k y_k x'_k y'_k]^T$, x_k, y_k are the object center in the x, y -axis location, x'_k, y'_k are the x, y axis speed. $Z_k = [x_{ck} y_{ck}]^T$ is the observation vector. P_k^- is apriori estimate error covariance and P_k is aposteriori estimate error covariance.

The search for object is carried in a region selected based on the estimated observation matrix Z_k . The Kalman filter tracker can handle variations in pose and occlusion. It fails if image is not properly illuminated.

B. Mean Shift Tracking

The mean shift algorithm searches the target in next frame around the location of that object in present frame. The target is represented using histogram bins [19]. Given the distribution $\hat{q}_u, u = 1, \dots, m$ of the face model and predicted location of face new state is derived using Algorithm 3. Mean shift tracking is computationally efficient. If inter-frame displacement is large and image contains objects that are similar to tracking object mean shift tracking fails.

VI. RESULTS

The skin color based methods are fast and satisfy the real-time constraints. The results of proposed methods are represented in Table II. Amongst YC_bC_r and HSV , YC_bC_r based skin color detection is fast and better. Eye detection using pixel count method requires more number of calculations as compared to integral projection method. Time required for

Algorithm 3 Mean Shift

1: Compute the distribution $(\hat{p}_u(\hat{y}_0)), u = 1, \dots, m$ and evaluate

$$\hat{\rho}(\hat{y}_0) = \sum_{u=1}^m \sqrt{\hat{p}_u(\hat{y}_0)} \hat{q}_u$$

2: Derive the weights $(w_i), i = 1, \dots, n_h$

3: Derive new location

$$\hat{y}_1 = \frac{\sum_{i=1}^{n_h} x_i w_i g\left(\left\|\frac{\hat{y}_0 - x_i}{h}\right\|^2\right)}{\sum_{i=1}^{n_h} w_i g\left(\left\|\frac{\hat{y}_0 - x_i}{h}\right\|^2\right)}$$

Update $(\hat{p}_u(\hat{y}_1)), u = 1, \dots, m$ and evaluate

$$\hat{\rho}(\hat{y}_1) = \sum_{u=1}^m \sqrt{\hat{p}_u(\hat{y}_1)} \hat{q}_u$$

4: **while** $\hat{\rho}(\hat{y}_1) < \hat{\rho}(\hat{y}_0)$ **do**

5: $\hat{y}_1 \leftarrow \frac{1}{2}(\hat{y}_0 + \hat{y}_1)$

6: **end while**

7: **if** $\|\hat{y}_1 - \hat{y}_0\| < \epsilon$ **then**

8: **Stop**

9: **else**

10: Set $\hat{y}_0 \leftarrow \hat{y}_1$ and go to step 1.

11: **end if**

eye detection by projection method is less than the pixel count method.

Detection of desired feature in each frame requires large time which is undesirable for real-time applications. The mean shift algorithm searches the target around the location of that feature in previous frame. The Kalman filter predicts the location of target in next frame. Tracking methods reduces the search area which minimizes computation time.

TABLE II. PERFORMANCE OF PROPOSED METHOD

Face	Eye	Tracking	Time(sec)		
			Video1	Video2	Video3
YCbCr	Pixel	Kalman	108.6	132.88	153.6
		Mean shift	25.22	38.39	55.1
	Projection	Kalman	100.34	105.13	139.44
		Mean shift	28.25	38.23	54.92
HSV	Pixel	Kalman	134.4	135.36	231.36
		Mean shift	31.24	42.23	71.34
	Projection	Kalman	120.7	121.44	192.1
		Mean shift	32.72	48.32	59.24

VII. CONCLUSIONS

The real-time tracking requires faster computations. Face detection methods based on intensity information are not so efficient and requires pre-processing. The color based methods for face detection are simpler and fast. Feature extraction based methods requires large computation and complex classifier to classify these features. The eye detection based on intensity variation is simpler and convenient approach. Eye detection methods are restricted to face region therefore performance of eye detection methods are dependent on the performance of the face detection methods. With the use of tracking algorithms like kalman filter and mean shift analysis, tracking time is reduced and meets the requirements of real-time applications. Kalman filter and mean shift tracking algorithms are popularly used for tracking.

Human tracking is necessity of future for security concern and can be used at places of work to keep eye on activities by the concerns.

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